

# Session 3 and 4 Summary

## Source links embedded in the EPUB

UNSAFE\_HTML\_20260623145058.html: [https://blackboard.ie.edu/bbcswebdav/pid-2762666-dt-content-rid-66158981\\_1/xid-66158981\\_1?Kq3cZcYS15=6671a827ba5c4c2183a4e13eef765cf0&VxJw3wfC56=1782578752&3cCnGYSz89=Eo2Y%2BB7yh2SAnwzvDSBxcsT2JSYf1J27vX8QLEjtUfQ%3D](https://blackboard.ie.edu/bbcswebdav/pid-2762666-dt-content-rid-66158981_1/xid-66158981_1?Kq3cZcYS15=6671a827ba5c4c2183a4e13eef765cf0&VxJw3wfC56=1782578752&3cCnGYSz89=Eo2Y%2BB7yh2SAnwzvDSBxcsT2JSYf1J27vX8QLEjtUfQ%3D)

## Class summary



# AI Challenge • Sessions 3 & 4 Summary

Model strategy, tokenomics, prompting, system roles, open models, and local AI

### WHY THESE SESSIONS MATTER

- Students learned to choose the right AI approach instead of using the biggest model for everything.
- The class connected cost, context, memory, hallucinations, and model choice to real company decisions.
- Prompting moved from casual chatting to structured design using role, system, and model strategy.

### SESSION 3 • MODEL STRATEGY & TOKENOMICS

**A**

#### Deterministic vs probabilistic

Traditional workflows are predictable and reliable. Generative AI is probabilistic and should be used only where variation is acceptable.

**B**

#### 3 company challenges

Memory limits, token cost, and hallucinations are the recurring problems companies must manage.

**C**

#### Context is everything

Better context improves answers, reduces hallucinations, and helps control cost.

**D**

#### Model-agnostic strategy

Build your process around context, safeguards, and orchestration. Swap models whenever a cheaper or better one appears.

**E**

#### Model routing

Use a smaller or cheaper model for simple tasks and a stronger model only for high-risk or complex tasks.

### TOKENOMICS EXPLAINED



Students used token counters to estimate cost per prompt and per future AI workflow.

### TOOLS DISCOVERED IN SESSION 3



#### OpenRouter

A gateway to many frontier models, useful for comparing options and routing requests.



#### Rankings & usage

Real-world developer and company usage can be more meaningful than vendor benchmark claims.



#### Pick the right model

Students completed an exercise selecting the best model for different business tasks.

### SESSION 4 • PROMPTING FOUNDATIONS

**A**

#### Three AI modes

Chat, Thinking, and Agentic AI each behave differently and require different prompting styles.

**B**

#### Chat mode

Fast and cheap. Best for direct answers and lightweight tasks.

**C**

#### Thinking mode

Uses more reasoning steps, often improving quality but increasing cost and latency.

**D**

#### Agentic AI

Uses tools, performs multi-step work, and is designed for longer independent execution.

**E**

#### Use only what you need

Do not enable heavy reasoning or expensive models when a simpler option is enough.

### PROMPTING TECHNIQUES



#### Role prompting

Assigning a role like marketing expert or investment advisor guides the model toward more relevant language and structure.



#### User prompt

The request the user types in the conversation.



#### System prompt

Persistent hidden instructions that define how the AI should behave.



#### Assistant behavior

The resulting tone, boundaries, and style emerge from the system setup plus the user request.

Students compared prompts with and without role framing and observed the quality difference.

### OPEN vs CLOSED MODELS



#### Open models

- Transparent weights
- Can be downloaded and run locally
- Useful for privacy, customization, and cost control



#### Closed / frontier models

- Usually strongest performance
- Cloud-based and easy to use
- Require trust in the provider and may cost more

Open source matters not only because it is cheaper, but because the weights and structure can be inspected more openly.

### HUGGING FACE & LOCAL AI



#### Hugging Face

A major hub for open models and AI spaces across text, image, audio, video, and more.



#### Spaces

Students explored ready-made demos and tools, including video-generation spaces.



#### Ollama

A practical way to run local models on your own machine.



#### Why local models matter

Privacy, zero cloud exposure, and lower cost for sensitive company workflows.

### CLASS ACTIVITIES



Deterministic vs probabilistic quiz



Token counter exercise



OpenRouter model exploration



Role prompting exercise



System prompt experiment



Hugging Face spaces demo



Homework: install Ollama and try a local model

### BIG TAKEAWAYS



Choose the right model, not the most expensive one.



Context design is central to quality, safety, and cost.



Prompting is a skill, not just typing requests.



Open models and local deployment are essential for privacy-sensitive use cases.



Real AI strategy is orchestration: task, model, context, tools, and cost working together.



## AI Challenge • Sessions 3 & 4 • Blackboard recap